

Proposal for Smart Adaptive Traffic Light Control System for a Four-Way Junction

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Abstract—This proposal presents an adaptive traffic control system for four-way intersections using an STM32 Cortex-M4 microcontroller and HC-SR04 ultrasonic sensors. The system adjusts green-light durations dynamically based on real-time traffic density to improve traffic flow and reduce congestion. A circular control sequence with bounded timing limits is implemented to ensure fair lane allocation and prevent vehicle starvation. Pedestrian push-button requests are also handled in a controlled manner to maintain safe and balanced junction operation. The proposed prototype aims to demonstrate improved traffic efficiency and reliable multidirectional traffic management.

Index Terms—STM32 Cortex-M4, Ultrasonic Sensor, Circular Sequence, Pedestrian

I. INTRODUCTION

Traditional four-way intersections rely on fixed timers that cannot adapt to changing traffic volumes. This rigidity creates bottlenecks, holding busy roads at red lights while empty streets receive green cycles [1].

However, managing traffic strictly by volume introduces new issues. Without constraints, low-traffic lanes can experience indefinite wait times (lane starvation) [2], and immediate pedestrian push-buttons can repeatedly interrupt and halt overall vehicle flow.

To resolve this, this project introduces an adaptive traffic controller powered by an STM32 microcontroller. By continuously processing data from ultrasonic sensors, the system dynamically adjusts green-light durations based on real-time density. It enforces maximum wait limits to prevent starvation and schedules pedestrian crossings efficiently, providing a robust upgrade over conventional traffic systems [3].

II. PROPOSED SOLUTION

A. Adaptive Priority Traffic Control

The proposed system uses an STM32 Cortex-M4 microcontroller to manage a four-way traffic intersection dynamically [4]. By collecting real-time traffic data, the system adjusts green light durations based on vehicle density to improve traffic flow and reduce congestion. The controller also manages countdown displays and pedestrian crossing requests safely.

HC-SR04 ultrasonic sensors are placed at different distances on each lane to detect traffic density more accurately [5]. This setup allows the system to identify whether a lane

has low traffic, heavy traffic, or no vehicles. Based on the detected traffic condition, the green light duration is adjusted dynamically.

In addition, pedestrian push buttons are included to allow safe road crossing. When a button is pressed, the system triggers a controlled interrupt to provide pedestrians with a suitable crossing time while minimizing disruption to vehicle traffic.

III. HARDWARE CHOICES

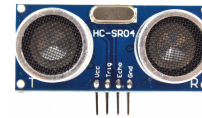


Fig. 1. HC-SR04 Ultrasonic Sensor



Fig. 2. Push Button

While a complete four-way system requires eight HC-SR04 sensors and eight push buttons, this prototype is simplified to use all eight sensors but only one push button to demonstrate a single pedestrian crossing.

IV. ADAPTIVE CONTROL STRATEGY

The proposed algorithm monitors traffic density for each lane (North, East, South, and West) using ultrasonic sensors. The system gives priority to lanes with higher traffic while also considering waiting time to prevent lane starvation.

Green light durations are adjusted dynamically based on real-time traffic conditions within predefined safety limits. After the maximum green time is reached, the system safely transitions through yellow and all-red states before switching lanes.

The algorithm also considers pedestrian push-button requests. When a request is detected, the system allocates a safe crossing period without causing excessive disruption to vehicle traffic. This approach helps improve traffic flow while maintaining fairness and safe pedestrian access.

V. BLOCK DIAGRAM AND FLOWCHART

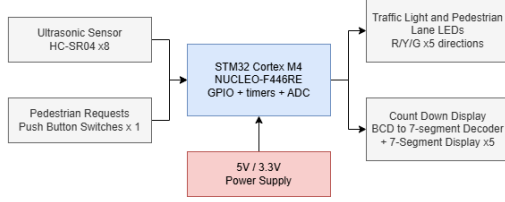


Fig. 3. STM32-Based Traffic Light Control System

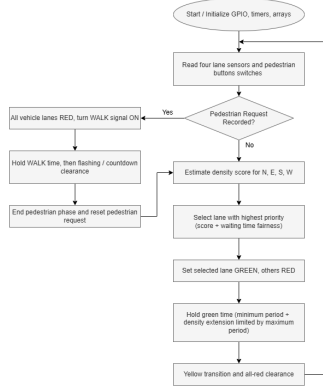


Fig. 4. Flowchart of Traffic Light Control System

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